

3.2 Plan: The molecular formula of sucrose gives the mole amount of each element in 1 mole of sucrose. Avogadro's number allows the change from moles to atoms.

Solution:

$$\text{a) Moles of C atoms} = (1 \text{ mol Sucrose}) \left(\frac{12 \text{ mol C}}{1 \text{ mol C}_{12}\text{H}_{22}\text{O}_{11}} \right) = \mathbf{12 \text{ mol C}}$$

$$\text{b) C atoms} = (2 \text{ mol C}_{12}\text{H}_{22}\text{O}_{11}) \left(\frac{12 \text{ mol C}}{1 \text{ mol C}_{12}\text{H}_{22}\text{O}_{11}} \right) \left(\frac{6.022 \times 10^{23} \text{ C atoms}}{1 \text{ mol C}} \right) = \mathbf{1.445 \times 10^{25} \text{ C atoms}}$$

3.8 Plan: Locate each of the elements on the periodic table and record its atomic mass. The atomic mass of the element times the number of atoms present in the formula gives the mass of that element in one mole of the substance. The molar mass is the sum of the masses of the elements in the substance.

Solution:

a) The molar mass, $\mathcal{M}$, is the sum of the atomic weights, expressed in g/mol:

$$\text{Sr} = (1 \text{ mol Sr}) (87.62 \text{ g Sr/mol Sr}) = 87.62 \text{ g Sr/mol Sr(OH)}_2$$

$$\text{O} = (2 \text{ mol O}) (16.00 \text{ g O/mol O}) = 32.00 \text{ g O/mol Sr(OH)}_2$$

$$\text{H} = (2 \text{ mol H}) (1.008 \text{ g H/mol H}) = \underline{2.016 \text{ g H/mol Sr(OH)}_2}$$

$$= \mathbf{121.64 \text{ g/mol of Sr(OH)}_2}$$

$$\text{b) } \mathcal{M} = (2 \text{ mol N}) (14.01 \text{ g N/mol N}) + (3 \text{ mol O}) (16.00 \text{ g O/mol O}) = \mathbf{76.02 \text{ g/mol of N}_2\text{O}_3}$$

$$\text{c) } \mathcal{M} = (1 \text{ mol Na}) (22.99 \text{ g Na/mol Na}) + (1 \text{ mol Cl}) (35.45 \text{ g Cl/mol Cl}) \\ + (3 \text{ mol O}) (16.00 \text{ g O/mol O}) = \mathbf{106.44 \text{ g/mol of NaClO}_3}$$

$$\text{d) } \mathcal{M} = (2 \text{ mol Cr}) (52.00 \text{ g Cr/mol Cr}) + (3 \text{ mol O}) (16.00 \text{ g O/mol O}) = \mathbf{152.00 \text{ g/mol of Cr}_2\text{O}_3}$$

3.12 Plan: The mass of a substance and its number of moles are related through the conversion factor of $\mathcal{M}$, the molar mass expressed in g/mol. The moles of a substance and the number of entities per mole are related by the conversion factor, Avogadro's number.

Solution:

$$\text{a) } \mathcal{M} \text{ of KMnO}_4 = 39.10 + 54.94 + (4 \times 16.00) = 158.04 \text{ g/mol of KMnO}_4$$

$$\text{Mass of KMnO}_4 = (0.68 \text{ mol KMnO}_4) \left(\frac{158.04 \text{ g KMnO}_4}{1 \text{ mol KMnO}_4} \right) = 107.47 = \mathbf{1.1 \times 10^2 \text{ g KMnO}_4}$$

$$\text{b) } \mathcal{M} \text{ of Ba(NO}_3)_2 = 137.3 + (2 \times 14.01) + (6 \times 16.00) = 261.3 \text{ g/mol Ba(NO}_3)_2$$

$$\text{Moles of O atoms} = (8.18 \text{ g Ba(NO}_3)_2) \left(\frac{1 \text{ mol Ba(NO}_3)_2}{261.3 \text{ g Ba(NO}_3)_2} \right) \left(\frac{6 \text{ mol O atoms}}{1 \text{ mol Ba(NO}_3)_2} \right) \\ = 0.18783 = \mathbf{0.188 \text{ mol O atoms}}$$

$$\text{c) } \mathcal{M} \text{ of CaSO}_4 \cdot 2\text{H}_2\text{O} = 40.08 + 32.07 + (6 \times 16.00) + (4 \times 1.008) = 172.18 \text{ g/mol}$$

(Note that the waters of hydration are included in the molar mass.)

$$\text{O atoms} = (7.3 \times 10^{-3} \text{ g Ca Cmpd}) \left(\frac{1 \text{ mol Ca Cmpd}}{172.18 \text{ g Ca Cmpd}} \right) \left(\frac{6 \text{ mol O atoms}}{1 \text{ mol Ca Cmpd}} \right) \left(\frac{6.022 \times 10^{23} \text{ O atoms}}{1 \text{ mol O atoms}} \right) \\ = 1.5319 \times 10^{20} = \mathbf{1.5 \times 10^{20} \text{ O atoms}}$$

- 3.16 Plan: The formula of each compound must be determined from its name. The molar mass for each formula comes from the formula and atomic masses from the periodic table. Avogadro's number is also necessary to find the number of particles.

Solution:

a) Carbonate is a polyatomic anion with the formula, CO_3^{2-} . Copper (I) indicates Cu^+ . The correct formula for this ionic compound is Cu_2CO_3 .

$$M \text{ of } \text{Cu}_2\text{CO}_3 = (2 \times 63.55) + 12.01 + (3 \times 16.00) = 187.11 \text{ g/mol}$$

$$\text{Mass } \text{Cu}_2\text{CO}_3 = (8.35 \text{ mol } \text{Cu}_2\text{CO}_3) \left(\frac{187.11 \text{ g } \text{Cu}_2\text{CO}_3}{1 \text{ mol } \text{Cu}_2\text{CO}_3} \right) = 1562.4 = \mathbf{1.56 \times 10^3 \text{ g } \text{Cu}_2\text{CO}_3}$$

b) Dinitrogen pentaoxide has the formula N_2O_5 . Di- indicates 2 N atoms and penta- indicates 5 O atoms.

$$M \text{ of } \text{N}_2\text{O}_5 = (2 \times 14.01) + (5 \times 16.00) = 108.02 \text{ g/mol}$$

$$\begin{aligned} \text{Mass } \text{N}_2\text{O}_5 &= (4.04 \times 10^{20} \text{ } \text{N}_2\text{O}_5 \text{ molecules}) \left(\frac{1 \text{ mol } \text{N}_2\text{O}_5}{6.022 \times 10^{23} \text{ } \text{N}_2\text{O}_5 \text{ molecules}} \right) \left(\frac{108.02 \text{ g } \text{N}_2\text{O}_5}{1 \text{ mol } \text{N}_2\text{O}_5} \right) \\ &= 0.072468 = \mathbf{0.0725 \text{ g } \text{N}_2\text{O}_5} \end{aligned}$$

c) The correct formula for this ionic compound is NaClO_4 ; Na has a charge of +1 (Group 1 ion) and the perchlorate ion is ClO_4^- . There are Avogadro's number of entities (in this case, formula units) in a mole of this compound.

$$M \text{ of } \text{NaClO}_4 = 22.99 + 35.45 + (4 \times 16.00) = 122.44 \text{ g/mol}$$

$$\text{Moles } \text{NaClO}_4 = (78.9 \text{ g } \text{NaClO}_4) \left(\frac{1 \text{ mol } \text{NaClO}_4}{122.44 \text{ g } \text{NaClO}_4} \right) = 0.644397 = \mathbf{0.644 \text{ mol } \text{NaClO}_4}$$

FU = formula units

$$\begin{aligned} \text{FU } \text{NaClO}_4 &= (78.9 \text{ g } \text{NaClO}_4) \left(\frac{1 \text{ mol } \text{NaClO}_4}{122.44 \text{ g } \text{NaClO}_4} \right) \left(\frac{6.022 \times 10^{23} \text{ FU } \text{NaClO}_4}{1 \text{ mol } \text{NaClO}_4} \right) = \\ &3.88056 \times 10^{23} = \mathbf{3.88 \times 10^{23} \text{ FU } \text{NaClO}_4} \end{aligned}$$

d) The number of ions or atoms is calculated from the formula units given in part c. Note the unrounded initially calculated value is used to avoid intermediate rounding.

$$3.88056 \times 10^{23} \text{ mol } \text{NaClO}_4 \left(\frac{1 \text{ Na}^+ \text{ ion}}{1 \text{ FU } \text{NaClO}_4} \right) = \mathbf{3.88 \times 10^{23} \text{ Na}^+ \text{ ions}}$$

$$3.88056 \times 10^{23} \text{ mol } \text{NaClO}_4 \left(\frac{1 \text{ ClO}_4^- \text{ ion}}{1 \text{ FU } \text{NaClO}_4} \right) = \mathbf{3.88 \times 10^{23} \text{ ClO}_4^- \text{ ions}}$$

$$3.88056 \times 10^{23} \text{ mol } \text{NaClO}_4 \left(\frac{1 \text{ Cl atom}}{1 \text{ FU } \text{NaClO}_4} \right) = \mathbf{3.88 \times 10^{23} \text{ Cl atoms}}$$

$$3.88056 \times 10^{23} \text{ mol } \text{NaClO}_4 \left(\frac{4 \text{ O atoms}}{1 \text{ FU } \text{NaClO}_4} \right) = \mathbf{1.55 \times 10^{24} \text{ O atoms}}$$

- 3.18 Plan: Determine the formula and the molar mass of each compound. The formula gives the number of atoms of each type of element present. Masses come from the periodic table. Mass percent = (total mass of element in the substance/molar mass of substance) x 100.

Solution:

a) Ammonium bicarbonate is an ionic compound consisting of ammonium ions, NH_4^+ and bicarbonate ions, HCO_3^- . The formula of the compound is NH_4HCO_3 .

$$\mathcal{M} \text{ of } \text{NH}_4\text{HCO}_3 = (14.01 \text{ g/mol}) + (5 \times 1.008 \text{ g/mol}) + (12.01 \text{ g/mol}) + (3 \times 16.00 \text{ g/mol}) \\ = 79.06 \text{ g/mol } \text{NH}_4\text{HCO}_3$$

In 1 mole of ammonium bicarbonate, with a mass of 79.06 g, there are 5 moles of H atoms with a mass of 5.040 g.

$$\frac{(5 \text{ mol H})(1.008 \text{ g/mol H})}{79.06 \text{ g/mol } \text{NH}_4\text{HCO}_3} \times 100\% = 6.374905 = \mathbf{6.375\% \text{ H}}$$

b) Sodium dihydrogen phosphate heptahydrate is a salt that consists of sodium ions, Na^+ , dihydrogen phosphate ions, H_2PO_4^- , and seven waters of hydration. The formula is $\text{NaH}_2\text{PO}_4 \cdot 7\text{H}_2\text{O}$. Note that the waters of hydration are included in the molar mass.

$$\mathcal{M} \text{ of } \text{NaH}_2\text{PO}_4 \cdot 7\text{H}_2\text{O} = (22.99 \text{ g/mol}) + (16 \times 1.008 \text{ g/mol}) + (30.97 \text{ g/mol}) + (11 \times 16.00 \text{ g/mol}) \\ = 246.09 \text{ g/mol } \text{NaH}_2\text{PO}_4 \cdot 7\text{H}_2\text{O}$$

In each mole of $\text{NaH}_2\text{PO}_4 \cdot 7\text{H}_2\text{O}$ (with mass of 246.09 g), there are $11 \times 16.00 \text{ g/mol}$ or 176.00 g of oxygen.

$$\frac{(11 \text{ mol O})(16.00 \text{ g/mol O})}{246.09 \text{ g/mol}} \times 100\% = 71.51855 = \mathbf{71.52\% \text{ O}}$$

- 3.23 Plan: Determine the formula of cisplatin from the figure, and then calculate the molar mass from the formula. The molar mass is necessary for the subsequent calculations.

Solution:

The formula for cisplatin is $\text{Pt}(\text{Cl})_2(\text{NH}_3)_2$

$$\mathcal{M} \text{ of } \text{Pt}(\text{Cl})_2(\text{NH}_3)_2 = 195.1 + (2 \times 35.45) + (2 \times 14.01) + (6 \times 1.008) = 300.1 \text{ g/mol}$$

$$\text{a) Moles cisplatin} = (285.3 \text{ g cisplatin}) \left(\frac{1 \text{ mol cisplatin}}{300.1 \text{ g cisplatin}} \right) = 0.9506831 = \mathbf{0.9507 \text{ mol cisplatin}}$$

$$\text{b) H atoms} = (0.98 \text{ mol cisplatin}) \left(\frac{6 \text{ mol H}}{1 \text{ mol cisplatin}} \right) \left(\frac{6.022 \times 10^{23} \text{ H atoms}}{1 \text{ mol H}} \right) \\ = 3.540936 \times 10^{24} = \mathbf{3.5 \times 10^{24} \text{ H atoms}}$$

- 3.28 Plan: The volume must be converted from cubic feet to cubic centimeters (or vice versa). The volume and the density will give you mass, and the mass with the molar mass gives you moles. Part (b) requires a conversion from cubic decimeters to cubic centimeters. The density allows you to change these cubic centimeters to mass, the molar mass allows you to find moles, and finally Avogadro's number allows you to make the last step.

Solution:

The molar mass of galena (PbS) is 239.3 g/mol.

$$\text{a) Moles PbS} = (1.00 \text{ ft}^3 \text{ PbS}) \left(\frac{(12 \text{ in})^3}{(1 \text{ ft})^3} \right) \left(\frac{(2.54 \text{ cm})^3}{(1 \text{ in})^3} \right) \left(\frac{7.46 \text{ g PbS}}{1 \text{ cm}^3} \right) \left(\frac{1 \text{ mol PbS}}{239.3 \text{ g PbS}} \right) = 882.7566886 = \mathbf{883 \text{ mol PbS}}$$

b) Lead atoms =

$$(1.00 \text{ dm}^3 \text{ PbS}) \left(\frac{(0.1 \text{ m})^3}{(1 \text{ dm})^3} \right) \left(\frac{(1 \text{ cm})^3}{(10^{-2} \text{ m})^3} \right) \left(\frac{7.46 \text{ g PbS}}{1 \text{ cm}^3} \right) \left(\frac{1 \text{ mol PbS}}{239.3 \text{ g PbS}} \right) \left(\frac{1 \text{ mol Pb}}{1 \text{ mol PbS}} \right) \left(\frac{6.022 \times 10^{23} \text{ Pb atoms}}{1 \text{ mol Pb}} \right) \\ = 1.87731 \times 10^{25} = \mathbf{1.88 \times 10^{25} \text{ Pb atoms}}$$

3.33 Plan: Examine the number of atoms of each type in the compound. Divide all atom numbers by any common factor. The final answers must be the lowest whole-number values.

Solution:

a) C_2H_4 has a ratio of 2 carbon atoms to 4 hydrogen atoms, or 2:4. This ratio can be reduced to 1:2, so that the empirical formula is **CH_2** . The empirical formula mass is $12.01 + 2(1.008) = \mathbf{14.03 \text{ g/mol}}$.

b) The ratio of atoms is 2:6:2, or 1:3:1. The empirical formula is **CH_3O** and its empirical formula mass is $12.01 + 3(1.008) + 16.00 = \mathbf{31.03 \text{ g/mol}}$.

c) Since, the ratio of elements cannot be further reduced, the molecular formula and empirical formula are the same, **N_2O_5** . The formula mass is $2(14.01) + 5(16.00) = \mathbf{108.02 \text{ g/mol}}$.

d) The ratio of elements is 3 atoms of barium to 2 atoms of phosphorus to 8 atoms of oxygen, or 3:2:8. This ratio cannot be further reduced, so the empirical formula is also **$Ba_3(PO_4)_2$** , with a formula mass of $3(137.3) + 2(30.97) + 8(16.00) = \mathbf{601.8 \text{ g/mol}}$.

e) The ratio of atoms is 4:16, or 1:4. The empirical formula is **TeI_4** , and the formula mass is $127.6 + 4(126.9) = \mathbf{635.2 \text{ g/mol}}$.

3.39 Plan: The percent oxygen is 100% minus the percent nitrogen. Assume 100 grams of sample, and then the moles of each element may be found. Divide each of the moles by the smaller value, and convert to whole numbers to get the empirical formula. The empirical formula mass and the given mass lead to the molecular formula.

Solution:

$$\text{a) } \% \text{ O} = 100\% - \% \text{ N} = 100\% - 30.45\% \text{ N} = 69.55\% \text{ O}$$

For a 100-gram sample, the mass, in grams, is numerically identical to the mass percent.

$$\text{Moles N} = (30.45 \text{ g N}) \left(\frac{1 \text{ mol N}}{14.01 \text{ g N}} \right) = 2.1734 \text{ mol N}$$

$$\text{Moles O} = (69.55 \text{ g O}) \left(\frac{1 \text{ mol O}}{16.00 \text{ g O}} \right) = 4.3469 \text{ mol O}$$

$$\left(\frac{2.1734 \text{ mol N}}{2.1734 \text{ mol N}} \right) = 1 \qquad \left(\frac{4.3469 \text{ mol O}}{2.1734 \text{ mol N}} \right) = 2$$

The empirical formula is **NO_2** .

$$\text{b) Formula mass of empirical formula} = 14.01 \text{ g N/mol} + 2(16.00 \text{ g O/mol}) = 46.01 \text{ g/mol}$$

$$\frac{\text{molar mass}}{\text{empirical formula mass}} = \frac{90 \text{ g/mol}}{46 \text{ g/mol}} = 2$$

Thus, the molecular formula is twice the empirical formula. Note: only an approximate value is needed. The molecular formula is $2(NO_2) = \mathbf{N_2O_4}$.

- 3.44 Plan: Assume 100 grams of cortisol so the percentages are numerically equivalent to the masses of each element. Convert each of the masses to moles by using the molar mass of each element involved. Divide all moles by the lowest number of moles and convert to whole numbers to determine the empirical formula. The empirical formula mass and the given molar mass will then relate the empirical formula to the molecular formula.

Solution:

$$\text{Moles C} = (69.6 \text{ g C}) \left(\frac{1 \text{ mol C}}{12.01 \text{ g C}} \right) = 5.7952 \text{ mol C}$$

$$\text{Moles H} = (8.34 \text{ g H}) \left(\frac{1 \text{ mol H}}{1.008 \text{ g H}} \right) = 8.2738 \text{ mol H}$$

$$\text{Moles O} = (22.1 \text{ g O}) \left(\frac{1 \text{ mol O}}{16.00 \text{ g O}} \right) = 1.38125 \text{ mol O}$$

$$\left(\frac{5.7952 \text{ mol C}}{1.38125 \text{ mol O}} \right) = 4.20 \quad \left(\frac{8.2738 \text{ mol H}}{1.38125 \text{ mol O}} \right) = 6.00 \quad \left(\frac{1.38125 \text{ mol O}}{1.38125 \text{ mol O}} \right) = 1.00$$

The carbon value is not close enough to a whole number to round the value. The smallest number that 4.20 may be multiplied by to get close to a whole number is 5. (You may wish to prove this to yourself.) All three ratios need to be multiplied by five to get the empirical formula of $\text{C}_{21}\text{H}_{30}\text{O}_5$.

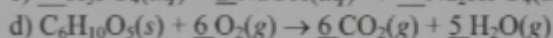
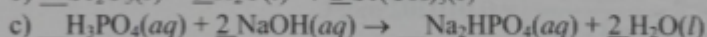
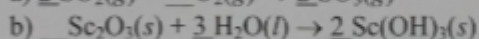
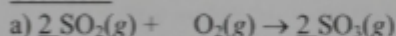
The empirical formula mass is:

$$21 (12.01 \text{ g C/mol}) + 30 (1.008 \text{ g H/mol}) + 5 (16.00 \text{ g O/mol}) = 362.45 \text{ g/mol}$$

The empirical formula mass and the molar mass given are the same, so the empirical and the molecular formulas are the same. The molecular formula is $\text{C}_{21}\text{H}_{30}\text{O}_5$.

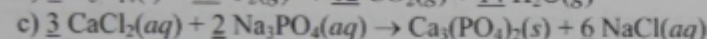
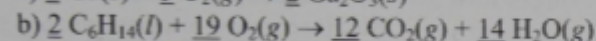
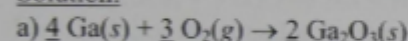
- 3.53 Plan: The balancing is a trial and error procedure. Do one blank/one element at a time.

Solution:



- 3.55 Plan: The names must first be converted to chemical formulas. The balancing is a trial and error procedure. Do one blank/one element at a time. Remember that oxygen is diatomic.

Solution:



- 3.63 Plan: Use the mole ratio from the balanced chemical equation to determine the moles produced. Use the moles with the molar mass to determine the grams produced.

Solution:

a) $\text{Moles Cl}_2 = (1.82 \text{ mol HCl}) \left(\frac{1 \text{ mol Cl}_2}{4 \text{ mol HCl}} \right) = 0.455 \text{ mol Cl}_2$

b) $\text{Grams Cl}_2 = (1.82 \text{ mol HCl}) \left(\frac{1 \text{ mol Cl}_2}{4 \text{ mol HCl}} \right) \left(\frac{70.90 \text{ g Cl}_2}{1 \text{ mol Cl}_2} \right) = 32.2595 = 32.3 \text{ g Cl}_2$

The beginning of the calculation is repeated to emphasize that the second part of the problem is simply an extension of the first part. There is no need to repeat the entire calculation, as only the final step times the answer of the first part will give the final answer to this part.

Hint: Always check to see if the initial equation is balanced. If the equation is not balanced, it should be balanced before proceeding.

- 3.73 **Plan:** Convert the given masses to moles and use the mole ratio from the balanced chemical equation to find the moles of CaO that will form. The reactant that produces the least moles of CaO is the limiting reactant. Convert the moles of CaO from the limiting reactant to grams using the molar mass.

Solution:

$$\text{a) Moles CaO from Ca} = (4.20 \text{ g Ca}) \left(\frac{1 \text{ mol Ca}}{40.08 \text{ g Ca}} \right) \left(\frac{2 \text{ mol CaO}}{2 \text{ mol Ca}} \right) = 0.104790 = \mathbf{0.105 \text{ mol CaO}}$$

$$\text{b) Moles CaO from O}_2 = (2.80 \text{ g O}_2) \left(\frac{1 \text{ mol O}_2}{32.00 \text{ g O}_2} \right) \left(\frac{2 \text{ mol CaO}}{1 \text{ mol O}_2} \right) = 0.17500 = \mathbf{0.175 \text{ mol CaO}}$$

c) **Calcium** is the limiting reactant since it will form less calcium oxide.

d) The mass of CaO formed is determined by the limiting reactant, Ca.

$$\text{Grams CaO} = (4.20 \text{ g Ca}) \left(\frac{1 \text{ mol Ca}}{40.08 \text{ g Ca}} \right) \left(\frac{2 \text{ mol CaO}}{2 \text{ mol Ca}} \right) \left(\frac{56.08 \text{ g CaO}}{1 \text{ mol CaO}} \right) = 5.8766 = \mathbf{5.88 \text{ g CaO}}$$

- 3.79 **Plan:** The question asks for the mass of each substance present at the end of the reaction. "Substance" refers to both reactants and products. Solve this problem using multiple steps. Recognizing that this is a limiting reactant problem, first write a balanced chemical equation. Using the molar relationships from the balanced equation, determine which reactant is limiting. Any product can be used to predict the limiting reactant; in this case, AlCl₃ is used. Additional significant figures are retained until the last step.

Solution:

The balanced chemical equation is:



Now determine the limiting reagent. We will use the moles of AlCl₃ produced to determine which is limiting.

$$\begin{aligned} \text{Mole AlCl}_3 \text{ from Al(NO}_2)_3 &= (72.5 \text{ g Al(NO}_2)_3) \left(\frac{1 \text{ mol Al(NO}_2)_3}{165.01 \text{ g Al(NO}_2)_3} \right) \left(\frac{1 \text{ mol AlCl}_3}{1 \text{ mol Al(NO}_2)_3} \right) \\ &= 0.439367 = 0.439 \text{ mol AlCl}_3 \end{aligned}$$

$$\text{Mole AlCl}_3 \text{ from NH}_4\text{Cl} = (58.6 \text{ g NH}_4\text{Cl}) \left(\frac{1 \text{ mol NH}_4\text{Cl}}{53.49 \text{ g NH}_4\text{Cl}} \right) \left(\frac{1 \text{ mol AlCl}_3}{3 \text{ mol NH}_4\text{Cl}} \right) = 0.365177 = 0.365 \text{ mol AlCl}_3$$

Ammonium chloride is the limiting reactant, and it is used for all subsequent calculations.

Mass of substances after the reaction:

Al(NO₂)₃:

Al(NO₂)₃ required to react with 58.6 g of NH₄Cl:

$$\begin{aligned} (58.6 \text{ g NH}_4\text{Cl}) \left(\frac{1 \text{ mol NH}_4\text{Cl}}{53.49 \text{ g NH}_4\text{Cl}} \right) \left(\frac{1 \text{ mol Al(NO}_2)_3}{3 \text{ mol NH}_4\text{Cl}} \right) \left(\frac{165.01 \text{ g Al(NO}_2)_3}{1 \text{ mol Al(NO}_2)_3} \right) \\ = 60.2579 = 60.3 \text{ g Al(NO}_2)_3 \end{aligned}$$

Al(NO₂)₃ remaining: 72.5 g – 60.3 g = **12.2 g Al(NO₂)₃**

NH₄Cl: **None left** since it is the limiting reagent.

AlCl₃:

$$(58.6 \text{ g NH}_4\text{Cl}) \left(\frac{1 \text{ mol NH}_4\text{Cl}}{53.49 \text{ g NH}_4\text{Cl}} \right) \left(\frac{1 \text{ mol AlCl}_3}{3 \text{ mol NH}_4\text{Cl}} \right) \left(\frac{133.33 \text{ g AlCl}_3}{1 \text{ mol AlCl}_3} \right) = 48.6891 = \mathbf{48.7 \text{ g AlCl}_3}$$

N₂:

$$(58.6 \text{ g NH}_4\text{Cl}) \left(\frac{1 \text{ mol NH}_4\text{Cl}}{53.49 \text{ g NH}_4\text{Cl}} \right) \left(\frac{3 \text{ mol N}_2}{3 \text{ mol NH}_4\text{Cl}} \right) \left(\frac{28.02 \text{ g N}_2}{1 \text{ mol N}_2} \right) = 30.697 = \mathbf{30.7 \text{ g N}_2}$$

H₂O:

$$(58.6 \text{ g NH}_4\text{Cl}) \left(\frac{1 \text{ mol NH}_4\text{Cl}}{53.49 \text{ g NH}_4\text{Cl}} \right) \left(\frac{6 \text{ mol H}_2\text{O}}{3 \text{ mol NH}_4\text{Cl}} \right) \left(\frac{18.02 \text{ g H}_2\text{O}}{1 \text{ mol H}_2\text{O}} \right) = 39.48297 = \mathbf{39.5 \text{ g H}_2\text{O}}$$

- 3.95 **Plan:** Remember that molarity is moles of solute in volume of solution.

Solution: Volumes may **not** be additive when two different solutions are mixed, so the final volume may be slightly different from 1000.0 mL. The correct method would state, "Take 100.0 mL of the 10.0 M solution and add water until the total volume is 1000 mL."

3.96 Plan: In all cases, the definition of molarity ($M = \frac{\text{moles solute}}{\text{L of solution}}$) will be important. Volume must be expressed in liters. The molar mass is used to convert moles to grams. The chemical formulas must be written to determine the molar mass.

Solution:

$$\text{a) Grams Ca(C}_2\text{H}_3\text{O}_2)_2 = (185.8 \text{ mL}) \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \left(\frac{0.267 \text{ mol Ca(C}_2\text{H}_3\text{O}_2)_2}{1 \text{ L}} \right) \left(\frac{158.17 \text{ g Ca(C}_2\text{H}_3\text{O}_2)_2}{1 \text{ mol Ca(C}_2\text{H}_3\text{O}_2)_2} \right) \\ = 7.84659 = \mathbf{7.85 \text{ g Ca(C}_2\text{H}_3\text{O}_2)_2}$$

$$\text{b) Moles KI} = (21.1 \text{ g KI}) \left(\frac{1 \text{ mol KI}}{166.0 \text{ g KI}} \right) = 0.127108 \text{ moles KI}$$

$$\text{Volume} = (500. \text{ mL}) \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) = 0.500 \text{ L}$$

$$\text{Molarity KI} = \frac{0.127108 \text{ mol KI}}{0.500 \text{ L}} = 0.254216 = \mathbf{0.254 \text{ M KI}}$$

$$\text{c) Moles NaCN} = \left(\frac{0.850 \text{ mol NaCN}}{1 \text{ L}} \right) (145.6 \text{ L}) = 123.76 = \mathbf{124 \text{ mol NaCN}}$$

3.100 Plan: These are dilution problems. Dilution problems can be solved by converting to moles and using the new volume, however, it is much easier to use $M_1V_1 = M_2V_2$. Part (c) may be done as two dilution problems or as a mole problem. The dilution equation does not require a volume in liters; it only requires that the volume units match.

Solution:

$$\text{a) } M_1 = 0.250 \text{ M KCl} \quad V_1 = 37.00 \text{ mL} \quad M_2 = ? \quad V_2 = 150.00 \text{ mL} \\ M_1V_1 = M_2V_2 \quad (0.250 \text{ M})(37.00 \text{ mL}) = (M_2)(150.0 \text{ mL}) \\ \frac{(0.250 \text{ M})(37.00 \text{ mL})}{150.0 \text{ mL}} = M_2 \quad M_2 = 0.061667 = \mathbf{0.0617 \text{ M KCl}}$$

$$\text{b) } M_1 = 0.0706 \text{ M (NH}_4)_2\text{SO}_4 \quad V_1 = 25.71 \text{ mL} \quad M_2 = ? \quad V_2 = 500.00 \text{ mL} \\ M_1V_1 = M_2V_2 \quad (0.0706 \text{ M})(25.71 \text{ mL}) = (M_2)(500.0 \text{ mL}) \\ \frac{(0.0706 \text{ M})(25.71 \text{ mL})}{500.0 \text{ mL}} = M_2 \quad M_2 = 0.003630 = \mathbf{0.00363 \text{ M (NH}_4)_2\text{SO}_4}$$

c) When working this as a mole problem it is necessary to find the individual number of moles of sodium ions in each separate solution. (Rounding to the proper number of significant figures will only be done for the final answer.)

$$\text{Moles Na}^+ \text{ from NaCl solution} = (3.58 \text{ mL}) \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \left(\frac{0.348 \text{ mol NaCl}}{1 \text{ L}} \right) \left(\frac{1 \text{ mol Na}^+}{1 \text{ mol NaCl}} \right) \\ = 0.00124584 \text{ mol Na}^+$$

$$\text{Moles Na}^+ \text{ from Na}_2\text{SO}_4 \text{ solution} = (500. \text{ mL}) \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \left(\frac{6.81 \times 10^{-2} \text{ mol Na}_2\text{SO}_4}{1 \text{ L}} \right) \left(\frac{2 \text{ mol Na}^+}{1 \text{ mol Na}_2\text{SO}_4} \right) \\ = 0.0681 \text{ mol Na}^+$$

$$\text{Molarity of Na}^+ = \left(\frac{(0.00124584 + 0.0681) \text{ mol Na}^+}{(3.58 + 500.) \text{ mL}} \right) \left(\frac{1 \text{ mL}}{10^{-3} \text{ L}} \right) = 0.1377058 = \mathbf{0.138 \text{ M Na}^+ \text{ ions}}$$

- 3.102 Plan: Use the density of the solution to find the mass of 1 L of solution. The 70.0% by mass translates to 70.0 g solute/100 g solution and is used to find the mass of HNO_3 in 1 L of solution. Convert mass of HNO_3 to moles to obtain moles/L, molarity.

Solution:

$$\text{a) Mass HNO}_3 \text{ per liter} = \left(\frac{1.41 \text{ g Solution}}{1 \text{ mL}} \right) \left(\frac{1 \text{ mL}}{10^{-3} \text{ L}} \right) \left(\frac{70.0 \text{ g HNO}_3}{100 \text{ g Solution}} \right) = \mathbf{987 \text{ g HNO}_3 / \text{L}}$$

$$\begin{aligned} \text{b) Molarity of HNO}_3 &= \left(\frac{1.41 \text{ g Solution}}{1 \text{ mL}} \right) \left(\frac{1 \text{ mL}}{10^{-3} \text{ L}} \right) \left(\frac{70.0 \text{ g HNO}_3}{100 \text{ g Solution}} \right) \left(\frac{1 \text{ mol HNO}_3}{63.02 \text{ g HNO}_3} \right) \\ &= 15.6617 = \mathbf{15.7 \text{ M HNO}_3} \end{aligned}$$

- 3.114 Plan: The moles of narceine and the moles of water are required. We can assume any mass of narceine hydrate (we will use 100 g), and use this mass to determine the mass of water present and convert the mass to moles of the hydrate. The mass of water will be converted to moles. Finally, the ratio of the moles of hydrate to moles of water will give the amount of water present.

Solution:

$$\begin{aligned} \text{Moles narceine hydrate} &= (100 \text{ g narceine hydrate}) \left(\frac{1 \text{ mol narceine hydrate}}{499.52 \text{ g narceine hydrate}} \right) \\ &= 0.20019 \text{ mol narceine hydrate} \end{aligned}$$

$$\begin{aligned} \text{Moles H}_2\text{O} &= (100 \text{ g narceine hydrate}) \left(\frac{10.8\% \text{ H}_2\text{O}}{100\% \text{ narceine hydrate}} \right) \left(\frac{1 \text{ mol H}_2\text{O}}{18.02 \text{ g H}_2\text{O}} \right) \\ &= 0.59933 \text{ mol H}_2\text{O} \end{aligned}$$

The ratio of water to hydrate is: $(0.59933 \text{ mol}) / (0.20019 \text{ mol}) = 3$

Thus, there are three water molecules per mole of hydrate. The formula for narceine hydrate is **narceine $\cdot$ 3H₂O**.

- 3.134 Plan: This problem may be done as two dilution problems with the two final molarities added, or, as done here, it may be done by calculating, then adding the moles and dividing by the total volume.

Solution:

$$\begin{aligned} M \text{ KBr} &= \frac{\text{Total Moles KBr}}{\text{Total Volume}} = \frac{\text{Moles KBr from Solution 1} + \text{Moles KBr from Solution 2}}{\text{Volume Solution 1} + \text{Volume Solution 2}} \\ M \text{ KBr} &= \frac{\left(\frac{0.053 \text{ mol KBr}}{1 \text{ L}} \right) (0.200 \text{ L}) + \left(\frac{0.078 \text{ mol KBr}}{1 \text{ L}} \right) (0.550 \text{ L})}{0.200 \text{ L} + 0.550 \text{ L}} = 0.071333 = \mathbf{0.071 \text{ M KBr}} \end{aligned}$$

- 3.161 Plan: Careful tracking of the units will carry you through this problem.

Solution:

$$\text{a) Dissolved salt} = (50.0 \text{ mL H}_2\text{O}) \left(\frac{10^{-3} \text{ L H}_2\text{O}}{1 \text{ mL H}_2\text{O}} \right) \left(\frac{2.50 \text{ kg Salt}}{1 \text{ L H}_2\text{O}} \right) \left(\frac{10^3 \text{ g}}{1 \text{ kg}} \right) = \mathbf{125 \text{ g salt}}$$

b) Convert the grams from part (a) to moles of salt, then to moles of cocaine, and finally to grams of cocaine. A calculation similar to that in part (a) will give the volume required.

$$\begin{aligned} \text{Mass cocaine} &= (125 \text{ g Salt}) \left(\frac{1 \text{ mol Salt}}{339.81 \text{ g Salt}} \right) \left(\frac{1 \text{ mol Cocaine}}{1 \text{ mol Salt}} \right) \left(\frac{303.35 \text{ g Cocaine}}{1 \text{ mol Cocaine}} \right) \\ &= 111.588 \text{ g cocaine (unrounded)} \end{aligned}$$

$$\begin{aligned} \text{Additional water needed} &= (111.588 \text{ g Cocaine}) \left(\frac{1 \text{ L}}{1.70 \text{ g Cocaine}} \right) - (50.0 \text{ mL}) \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \\ &= 65.59 = \mathbf{65.6 \text{ L H}_2\text{O}} \end{aligned}$$